INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

Department : Computer Science and Engineering 20

Invigilator's Signature

Total No. of Pages

Mid Semester ✓ End Semester Make- up Suppl. Exam

STUDENT'S NAME NISHIT PARMAR. Roll No.: CS26btech11049

Programme : BTech (BTech./ BDes /MSc/MTech/MDes/MA/PhD etc)

Course No.: CS1010 Course Title : Discreeb Maths for CS PART :

Question No.	1	2	3	4	5	6	7	8	9	10	Total
MARKS	3	0	2	2	1	2	2	8			

Q8) (a) True ✓

(b) False ✓

(c) False ✓

(d) True ✓

(e) False ✓

(f) False ✓

(g) True ✓

(h) True ✓

Q1) $\phi = (p \rightarrow (q \wedge r)) \vee (q \leftrightarrow r)$

(a) $(\neg p \vee (q \wedge r)) \vee (q \leftrightarrow r)$

$(\neg p \vee (q \wedge r)) \vee ((q \rightarrow r) \wedge (r \rightarrow q))$

$(\neg p \vee (q \wedge r)) \vee ((\neg q \vee r) \wedge (\neg r \vee q))$

$(\neg p) \vee ((q \wedge r) \vee (\neg q \wedge \neg r))$

$\neg p \vee (q \leftrightarrow r)$

(b) $(\neg p \vee (q \vee r)) \wedge ((q \wedge r) \vee (\neg q \wedge \neg r))$

$(\neg p) \vee (q \wedge r) \vee (q \vee r) \wedge (\neg q \vee \neg r)$ [C.N.F]

$\neg p \vee (q \wedge r) \vee (\neg (q \vee r))$

$((q \wedge r) \vee \neg (q \vee r)) \vee \neg p$

$\neg \left(p \wedge (\neg q \vee \neg r) \wedge (q \vee r) \right)$ [D.N.F]

Q2) $C(x,y)$: researcher x has collaborated with researcher y

(a) (ii) $\forall x, \exists y, (x \neq y), C(x,y)$

(b) (i) $\exists y, \forall x, C(x,y)$

Q3) a & b are odd integers

i.e $a = 2m+1$, $m \in \mathbb{Z}$

$b = 2n+1$, $n \in \mathbb{Z}$

(a) $(2m+1)^2 - (2n+1)^2 = (2m+1-2n-1)((2m+1)+(2n+1))$

$= 2(m-n) \cdot 2(m+n+1)$

$= \underline{4(m-n)(m+n+1)}$

Now, $(m-n)(m+n+1)$ is even in all cases, whether m & n are odd or even

So, $a^2-b^2 = 8K$, for some K,

(i) $(m = \text{even}, n = \text{even}) = (\text{even}) \cdot (\text{odd}) = (\text{even})$

$(m = \text{odd}, n = \text{odd}) = (\text{even}) \cdot (\text{odd}) = (\text{even})$

$(m = \text{even}, n = \text{odd}) = (\text{odd}) \cdot (\text{even}) = (\text{even})$

(ii) $(m = \text{odd}, n = \text{even}) = (\text{odd}) \cdot (\text{even}) = (\text{even})$

So, $a^2-b^2 = 8K$, for some K

and hence a^2-b^2 is divisible by 8.

(b) Converse:- if a number is divisible by 8, it is always of the form (a^2-b^2), for some values of a & b.

✗ Wrong converse.

In the proof above, we can take, $2K = (m-n)(m+n+1)$

m and n were not specific or special, so it is valid for any $m, n \in \mathbb{Z}$

So, we can prove as above in the reverse direction and thus the converse is also true.

Q6) (a) Consider, $A = \mathbb{Q}$ = set of all rational numbers

$B = \mathbb{Z}$ = set of all integers

Both A and B are countably infinite.

$A \cap B = \mathbb{Q} \cap \mathbb{Z}$ = set of rationals that are integers

So, $A \cap B$ is also countably infinite

(b) Consider, $A = \{0, -1, -2, \ldots\}$ = set of all negative integers including zero.

$B = \mathbb{N} = \{0, 1, 2, \ldots\}$ = set of all positive integers including zero.

Both A & B are countably infinite.

$A \cap B = \{0\}$

and $A \cap B$ is finite.

Q7)

Q2) (a) (i) $\exists x, \forall y, (x \neq y) \wedge C(x,y)$

(ii) $\exists y, \forall x, (x \neq y) \rightarrow C(x,y)$

(b) $\exists x, \forall y, \neg C(x,y)$

(i) No researcher has collaborated with every other researcher.

(ii) $\forall y, \exists x, \neg C(x,y)$

Every researcher has not collaborated with atleast one researcher.

Q4) Let $S_n(n)$ denote the machine, where n_n is the input and, $S_n : \mathbb{N} \mapsto \{L, U\}$

L: Lucky Number, U: Unlucky Number.

Let X denote the set of all such machines. and consider it to be countable.

So, we can list the machines

	1	2	3	4	—
S_1	:	.	.	.	
S_2	:				
S_3	:				
S_4					
S_5					
S_6					
S_7					
⋮					
)					

6

Here, each machine gives some output for a natural number.

Now, consider a new machine such that its output is the opposite of what $S_n(n)$ gives starting from $n=1$, along the moving along the diagonal.

This machine will have outputs diff than all the machines and thus be a new machine that is not in the set X.

So, our assumption that X is countable is incorrect.

Hence, X is uncountable set.

Q5) $|S| = |\mathbb{N}|$

Even though S seems to be the power set of $\mathbb{N}$, both have same cardinality.

How?